Thinking Skills and Creativity 38 (2020) 100750

Contents lists available at ScienceDirect

Thinking Skills and Creativity

journal homepage: www.elsevier.com/locate/tsc

Meta-analysis of interventions and their effectiveness in students' scientific creativity☆

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ARTICLE INFO

Keywords:
Meta-analysis
Scientific creativity
Interventions
Effectiveness
Creativity

ABSTRACT

As the importance of students' scientific creativity garners increasing attention, many interventions have been designed to cultivate such creativity. To date, however, the available evidence regarding the effectiveness of such interventions has not been systematically reviewed. In this meta-analysis, the results of interventions to improve students' scientific creativity have been brought together to determine which is the most effective. The analysis is based on 20 outcomes from 17 studies published between 1992 and 2019. Four types of intervention were systematically reviewed: *problem-solving*, *collaborative learning*, *conceptual construction*, and *scientific reasoning*. The findings demonstrate that *problem-solving* has a large effect on students' scientific creativity ($g = 1.54; n = 8$), as does *scientific reasoning* ($g = .89; n = 3$), while *collaborative learning* has a medium effect ($g = 0.76, n = 6$) and *conceptual construction* has a small effect ($g = .20; n = 3$). In conclusion, this paper found that interventions aimed at cultivating the product dimension of scientific creativity might be most effective in terms of students' scientific creativity, followed by those based on training traits and process dimension, in that order.

1. Introduction

Creativity is an indispensable ability of human beings (Akyıldız & Çelik, 2020; Huang, Peng, Chen, Tseng, & Hsu, 2017; Sternberg, 1999), and most research on it has been conducted since 1950s (Runco & Jaeger, 2012; Sternberg, 2006). Among these studies, creativity research connected with education has undoubtedly had the greatest and most enduring social impact (Guilford, 1970; Spendlove, 2008; Willerson & Mullet, 2017). Such research has focused on different aspects of creativity, including students' creativity in science education (Kind & Kind, 2007), which has received increasing attention (Huang et al., 2017; Kind & Kind, 2007; Yang, Hong, Lee, & Lin, 2019). Simultaneously, the importance of students' scientific creativity has been explicitly recommended in national science education documents (e.g., Council, 2012; Department for Education, 2013) and government policy (e.g., Ministry of Education, 2002). Therefore, science education must concern itself with students' scientific creativity and many interventions be designed to cultivate this skill. To date, there has been no systematic review of the evidence regarding the effectiveness of these interventions aimed at improving students' scientific creativity, and such research may serve as an important step for stimulating students' potential in scientific creativity. Therefore, this study attempts to integrate empirical research aimed to cultivate students' scientific creativity to create generalizations about the effectiveness of these interventions.

☆ This document is the results of the research project funded by the Educational Science in Shandong of 13th Five-Year Plan (YZ2017029).
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https://doi.org/10.1016/j.tsc.2020.100750
Received 2 November 2019; Received in revised form 10 August 2020; Accepted 18 October 2020
Available online 24 October 2020

In order to explore the effectiveness of interventions aimed to cultivate students' scientific creativity, the definition of such creativity is first reviewed through examination of previous research. Guilford (1950) introduced "divergent thinking" as a key concept to initially describe the creative process. In fact, the creative process includes not only the process of thinking in various directions in order to arrive at alternative solutions to a problem (divergent thinking) but also the process of thinking logically to arrive at one correct solution, which is described as "convergent thinking" (Guilford, 1967; Kind & Kind, 2007; Webb, Little, Cropper, & Roze, 2017; Zmigrod, Colzato, & Hommel, 2015). These two types of thinking constitute a complete creative process. Scientific creativity in particular leads to creative scientific work (Goldschmidt, 2016; Ramankulov, Usembayeva, Berdi, Omarov, & Shektibayev, 2016). In order to fruitfully understand this creative process, Campbell (1960) conceptualized it as a twofold procedure of blind variation (BV) and selective retention (SR), or BVSR, in which the BV component represents idea generation of divergent thinking and the SR component represents the process of convergent thinking (Jung, Mead, Carrasco, & Flores, 2013). BVSR, unfortunately, has not been developed into a full-fledged psychological theory of creativity (Simonton, 2011), but it provided exploratory work for the later research about mental representations and processes underlying scientific creativity. Simonton (2003, 2009, 2010, 2015) provided a theory using constrained stochastic behavior to analyze scientific creativity, and BVSR was one of the four assumptions of stochastic explanations of this theory. In this theory, the process of scientific creativity is regarded as constrained stochastic behaviors of scientists' science knowledge, which plays a central role in producing scientific work that is both novel and appropriate. It should be noticed that adolescent and mature scientists follow the same scientific creative process to create creative scientific work (Hu & Adey, 2002) even if the scientists are the developers of science knowledge and the students are learners of scientific knowledge (Kind & Kind, 2007). Therefore, students' scientific creativity can also be regarded as constrained stochastic behaviors of science knowledge, which plays a central role in their creative process. In fact, the importance of science knowledge in scientific creativity has been emphasized by many studies (Acar & van den Ende, 2016; Kocabas, 1993; Zulkarnaen, Supardi, & Jatmiko, 2018).

According to a description of students' creative process of scientific creativity, it can be concluded that the interventions that might lead to the accumulation of science knowledge (meaning the scientific knowledge that participates in quasi-random recombinations increases) or lead to higher quantities of the possible combinations of scientific knowledge (the probability of obtaining original and useful permutations increase) will cultivate students' scientific creativity (Simonton, 2003; Turnbull, Littlejohn, & Allan, 2010). In fact, there are two categories of interventions in this meta-analysis that cultivate students' scientific creativity in this way, ***collaborative learning*** and ***conceptual construction***.

Though various individuals follow the same creative process mentioned above, the traits of scientific creativity in this process vary with each individual. Fluency (the number of relevant ideas), flexibility (the categories covered by the responses given), and originality (the number of statistically infrequent ideas) have been regarded as central features of creativity by several researchers (Hu & Adey, 2002; Kharkhurin, 2017; Kim, 2006; Torrance, 1966), who also formed the trait dimensions of students' scientific creativity found in Hu and Adey's (2002) Scientific Structure Creativity Model (SSCM). It should be noted that these features are simply features of divergent thinking; as for convergent thinking, it also focuses on the integrative, original synthesis (Cropley, 2006; Lubart, 2016; Vidler, 1974). Therefore, in the present study, the features of divergent thinking and convergent thinking are used to describe the traits of scientific creativity of students represented in their scientific creative process. Of course, students' scientific creativity can also be improved through traits training, and in this study, a category of intervention called ***scientific reasoning*** cultivates students' scientific creativity in this way.

In addition to the process and trait dimension, several researchers have noted that all creative processes move toward the generation of certain products (Sak & Ayas, 2013; Simonton, 2003; Zeng, Proctor, & Salvendy, 2011). With regard to scientific creativity, these products refer to technical products, advances in science knowledge, understanding of scientific phenomena, and scientific problem-solving (Hu & Adey, 2002). Obviously, if an intervention can provide opportunities to help students understand scientific phenomena and improve problem-solving ability, then students' scientific creativity might be cultivated. In this study, a category of intervention named ***problem-solving*** improves students' scientific creativity in this way.

As mentioned above, in the present study, students' scientific creativity refers to the constrained stochastic behaviors of science knowledge, which produce ***product*** and can be described by personality ***traits***. The design of interventions that aim to cultivate it are rooted in one or more of its certain dimensions (Demirhan & Şahin, 2019; Lin, Hu, Adey, & Shen, 2003; Siew & Ambo, 2018). For example, some interventions cultivate students' scientific creativity from the dimension named ***product***, some from ***process*** dimension, and so on. These interventions all intend to improve students' scientific creativity, but evidence of their effectiveness has not been systematically evaluated, as was mentioned above. In order to complete this systematic review, this study reviewed and summarized all interventions aimed to cultivated students' scientific creativity, then the meta-analysis is used to determine if receiving the interventions mentioned in existing studies will improve students' scientific creativity based on studies with a quasi-experimental or truly experimental design. In addition, this study also examines which types of intervention most effectively develop students' scientific creativity and what the characteristics of the most effective intervention are.

2. Methods

2.1. Eligibility and exclusion criteria

Since the modern research on creativity only commenced in 1950, only studies published after 1-1-1950 were included in this study. This meta-analysis aims to systematically review the evidence on the effectiveness of interventions to improve students' scientific creativity; therefore, only studies that provide evidence of the cause-and-effect relationship between interventions and students' scientific creativity are included. While studies with a true experimental design can certainly do this, as McMillan and

Schumacher (2010) stated, in education, it is often difficult to meet the conditions of random assignment and manipulation. Therefore, in education, quasi-experimental design is often regarded as an alternative for true experimental design (Fraenkel, Wallen, & Hyun, 2011; McMillan & Schumacher, 2010).

Though single-subject design allows reasonable cause-and-effect conclusions (McMillan & Schumacher, 2010), it creates too much risk of bias regarding interventions (Thompson & Panacek, 2006). Therefore, such studies were excluded from this meta-analysis (McMillan & Schumacher, 2010).

Finally, because the study authors come from China, only studies written in English or Chinese were included. The eligibility and exclusion criteria can be summarized as follows:

Eligibility criteria:

1. Studies published after 1-1-1950.
2. Studies written in English or Chinese.
3. Only empirical studies investigating the quantitative effects of interventions (to improve students' scientific creativity).
4. Studies that set the dependent variable as students' scientific creativity were considered for the subsequent analysis.
5. Studies with quasi-experimental design (non-equivalent-group posttest design, non-equivalent-group pretest-posttest design) or true-experiment design (randomized equivalent group posttest and randomized equivalent group pretest-posttest design) (Sung, Lee, Yang, & Chang, 2019).
6. Studies with a full text available via university library systems or the internet.
7. Studies reporting sufficient statistical information to calculate an effect size.

Exclusion criteria:

1. Studies focused on samples of children with learning difficulties and disabilities.
2. Studies with pre-experiment design (one-group pretest-posttest design).
3. Correlation, survey, single-subject, and qualitative research that did not use an experimental design.

2.2. Information sources

To minimize bias, systematic reviews of interventions require a thorough, objective, and reproducible search of a range of sources to identify as many relevant studies as possible (Higgins et al., 2019). Generally, electronic databases are one of the primary search tools in modern meta-analysis due to their wide coverage (Card, 2015). However, use of electronic databases raises an issue regarding how to avoid publication bias. As noted by Card (2015), the extent to which one can avoid publication bias in a meta-analysis depends on the inclusion of gray literature (including academic theses, organization reports, government papers, etc.; Bernes et al., 2013), which may prove highly influential in syntheses, despite not being formally published. Therefore, the electronic databases selected in this study were ones that provide gray literature: ProQuest, ERIC, Web of science, PsycINFO, and Google scholar were searched for eligible studies from 1950 to 31-12-2019. Of these, ERIC contains journal and non-journal articles and reports obtained since 1966 (James & McMillan, 2013). Google Scholar includes approximately 100 million records of both academic and gray literature and it has received considerable attention as a method for searching for gray literature (Haddaway, Collins, Coughlin, & Kirk, 2015; Orduña-Malea, Ayllón, Martín-Martín, & López-Cózar, 2014). Web of Science (WOS) can provide not only academic but also gray literature (proceedings papers, etc.). ProQuest provides unpublished dissertations, proceedings papers, etc. PsycINFO provides not only academic but also gray literature (proceedings papers, etc.).

2.3. Search strategy

The search process was as follows:ProQuestAdvanced search was selected and ***Anywhere*** was set as *"scientif* creativ*"*. The settings were: publication date, 1-1-1950 to 12-31-2019; only English and Chinese written articles; and all document types (*Article*, *Book*, *Dissertation/Thesis*, etc.).ERIC*scientific AND creativity* was input in the ERIC search box.Web of ScienceThe Social Science Citation Index (SSCI) database and Science Citation Index Expand (SCI-Expanded) database were first selected. Then, the ***Topic*** was set to *"scientif* creativ*"*, ***Timespan*** as 1950 to 2019, and language as only English and Chinese written articles. All document types (*ARTICLE*, *BOOK REVIEW*, *PROCEEDINGS PAPER*, etc.) were selected.PsycINFOThe APA PsycArticles, Psychology and Behavioral Sciences Collection, and Academic Search Complete were selected. Then, the ***TX All Text*** was set as *"scientif* creativ*"*. The ***Published date*** was set from January-1950 to December-2019. Only English and Chinese written articles were selected. All document types (*Abstract*, *Article*, *Proceeding* etc.) were selected.Google scholar*allintitle:scientific creativity* was input in the search box and patents and citations were excluded. The ***year*** was set as 1950 to 2019.

2.4. Study selection

Microsoft Excel was used to manage the search results. First, the Google scholar retrieval results were downloaded by *Publish or perish 6* (Harzing, 2007) in the form of a sheet. The retrieval results from other databases were also downloaded in sheet form. Then, all sheets were joined together and duplicate files were hand-removed using Microsoft Excel.

Two authors checked articles for inclusion eligibility. Articles were initially screened for eligibility based on titles and abstracts.

The results from Google scholar were filtered for written language (English or Chinese only); those from other databases had been pre-filtered for language.

The search records were filtered using the inclusion and exclusion criteria as follows:

1. If a study's topic is clearly not about students' scientific creativity, it is excluded.
2. Second, two researchers labelled each study "T(TRUE)," "F(FALSE)," or "N(Not sure)" to show whether they satisfied the third and fourth eligibility criteria.
3. Third, the remaining eligibility criteria (fifth, sixth, seventh) and exclusion criteria were applied to the records retained after the second step.
4. The three steps above could be conducted based on the title and abstract. However, the second and third steps needed to be applied to the full-text copy of studies retained after the second and third steps iteratively.

In each step, agreement between reviewers was determined by calculating the Cohen's kappa score (Orwin & Vevea, 2009). Kappa values between 0.41 and 0.60 reflect moderate agreement, 0.61–0.80 reflect substantial agreement, while 0.81 or more reflect almost perfect agreement (Landis & Koch, 1977). Disagreements between reviewers were resolved by discussion.

2.5. Data collection process

Data extraction was conducted by two authors and disagreements were resolved by discussion. No data extraction piloting was conducted.

2.6. Data items

As the PRISMA for systematic review protocols (PRISMA-P) suggested (Moher et al., 2015), extracted data include study details (author, publication year, country), participants (number, percentage male/female, grade, number of experimental and control groups), intervention (theoretical basis, duration), compactors (treatment of experimental and control groups), outcomes (mean and standard deviation of control group and experimental group, or other statistical methods and their results), and measurement instruments.

2.7. Risk of bias in individual studies

Two reviewers independently assessed the risk of bias in each study, scored using the Cochrane risk of bias tool (Higgins et al., 2019, see Chapter 8, Tables 8.5.a and 8.5.d). The item on blinding of participants and personnel was split and scored for each of participants and personnel. Inter-rater agreement was measured by Cohen's kappa coefficient (95% confidence intervals). Disagreements were resolved through discussion. The results were presented in a risk of bias summary describing the review authors' judgments regarding risk of bias for each included study, and a risk of bias graph describing judgments were presented as percentages across all studies. The risk of bias was then discussed (Higgins et al., 2019). Many studies did not provide sufficient information to judge the bias risk of different dimensions. For example, it is difficult to meet the blinding of personnel condition (in fact, no study in this meta-analysis met this condition); therefore, it made no sense to label bias risk "unclear" in such instances. If a study did not provide sufficient information to judge the bias risk of a dimension, that dimension was labeled "high risk of bias(YES)."

2.8. Summary measures

As mentioned in Section 2.1, only studies with quasi-experimental or true-experiment design were included. Therefore, data were sought for all outcomes that describe the difference in students' scientific creativity between experimental and control groups after treatment. Of course, different statistical methods were used to describe these differences; for example, some studies provided the mean value and standard deviation of the control and experimental groups, some used ANCOVA to analyze the difference, and some used *t*-test.

Outcomes were prioritized as follows:

1. If the difference between control and experimental groups was examined before treatment and there was no significant difference between them in students' scientific creativity, these outcomes were ordered as: $\text{mean value and standard deviation} > \text{ANCOVA} > t$.
2. If the difference between control and experimental groups was not examined before treatment or there was no evidence to confirm no significant difference between groups in students' scientific creativity, outcomes were ordered as: $\text{ANCOVA} > \text{mean value and standard deviation} > t$.
3. If the mean value and standard deviation of the control and experimental groups or the result of ANCOVA were not provided, the *t*-value was recorded.

Furthermore, some studies only provided the mean value and standard deviation of the control and experimental groups for each component of scientific creativity (e.g., unusual uses, problem finding, etc.; see Lin et al., 2003). Thus, the mean value and standard deviation of each sub-item score were combined using the formula for combining groups (Higgins et al., 2019; for formula, see

Table 7.7.a) and the mean value and standard deviation of the two groups were then recorded.

After the outcomes of all studies had been recorded, the effect sizes (which provide a standardized measure of treatment effects, see Borenstein, Cooper, Hedges, & Valentine, 2009) of these outcomes were calculated. Effect sizes are typically calculated with Cohen's d or Hedges' g (Cohen, 2013; Hedges & Olkin, 2014), and produce similar results, but Hedges' *g* corrects for smaller sample sizes (e.g., less than 20) (Jeong, Hmelo-Silver, & Jo, 2019). To reduce biases from smaller samples, which is not uncommon in educational research, this study used the same effect size as Jeong et al. (2019), and calculated Hedges' g values of each study using R package esc (Lüdecke, 2019). The benchmarks of Hedges' g were presented by Cohen (2013), with 0.20 indicating a small effect, 0.50 a medium effect, and over 0.80 a large effect (Borman & Grigg, 2009).

2.9. Synthesis of results

Our criteria allowed for inclusion of different (combinations of) interventions in both the experimental and control groups; therefore, intervention diversity was expected in the comparisons made in the primary studies. As various interventions aimed at cultivating students' scientific creativity were included, an issue emerged regarding the appropriateness of combining the numerical results of all, or perhaps some, of the studies (Higgins et al., 2019). This issue was resolved using meta-analysis to summarize the results of independent studies (Hedges & Cooper, 2009).

This study aimed to analyze the effects of interventions to cultivate students' scientific creativity; therefore, if more than one study used similar experimental groups (using the same intervention), investigating the same outcome (all studies concern the cultivation of students' scientific creativity), quantitative synthesis using meta-analysis was considered. In this circumstance, if two interventions differed in name but used the same theoretical framework or affected students' scientific creativity in the same way (problem-solving, collaborative learning, etc.), then they were regarded as the same intervention. For example, though Siew, Chin, and Sombuling (2017) used *problem-based learning* to describe the interventions used in his studies, in fact, the intervention (Siew et al., 2017) provided the opportunity to enhance the contact between individuals which lead to the accumulation of science knowledge and the increment of the quantity of the possible combination of science knowledge. Therefore, this intervention was categorized as ***Collaborative learning*** rather than ***Problem-solving***. Two reviewers independently assigned the different interventions as problem-solving, collaborative learning, conceptual construction, or scientific reasoning. Inter-rater agreement was measured using Cohen's kappa coefficient (95% confidence intervals) and disagreements were resolved through discussion.

If the data of several studies were appropriate for synthesis, then these results were summarized using R package named meta (Balduzzi, Rücker, & Schwarzer, 2019). As it was not assumed that the estimated effects from the component studies came from a single homogeneous population (Schwarzer, Carpenter, & Rücker, 2015), the random-effects model was used to synthesize the results, estimated by the inverse variance method (Balduzzi et al., 2019; Higgins et al., 2019). Several issues should be considered in results synthesis:Unit of analysis issuesIn some studies, multiple experimental groups were compared to one control group. As Donker, de Boer, Kostons, van Ewijk, and van der Werf (2014) suggested, the number of students in the control group was divided by the number of experimental groups.Assessment of heterogeneityTreatment heterogeneity was tested by considering the variability in participant factors among trials (country, age, and gender) and trial factors (self-designed measurement instruments, duration). Statistical heterogeneity was tested using the χ^2 test (significance level: 0.01) and I^2 statistic (mild heterogeneity might account for less than 30% of the variability in point estimates, and notable heterogeneity substantially more than 50%, with 30% to 50% indicating moderate heterogeneity; see Higgins & Thompson, 2002). If high levels of heterogeneity exist among trials ($I^2 \geqslant 50\%$ or $P < 0.01$, Higgins et al., 2019), the participant factors among trials and trial factors were analyzed. The source of heterogeneity was explained through meta-regression.Dealing with missing dataWhen data were missing, the original study authors were contacted. If missing data could still not be obtained, only the available data were analyzed (Higgins et al., 2019).

After the above analysis, the power of the meta-analysis was calculated using the tool Quintana and Tiebel (2017) provided.

2.10. Publication bias

The published scientific papers only comprise a proportion of all the research conducted. Further, there is evidence to suggest that unpublished research may differ systematically from published research, since selectivity may exist in deciding what to publish (Song, Easterwood, Gilbody, Duley, & Sutton, 2000). There is also evidence that studies with significant outcomes are published more quickly than those with nonsignificant outcomes (Stern & Simes, 1997). If only published articles are used for meta-analysis, the results might be biased (Sutton, 2009), posing a major threat to meta-analytic validity, including the present research.

The funnel plot is generally considered a good exploratory tool for investigating publication bias and, like the more commonly used forest plot, for a visual summary of a meta-analytic dataset (Sutton, 2009). If there is no publication bias, the funnel plot is symmetric; otherwise, it is asymmetric. To circumnavigate certain issues related to the subjectivity of assessment of funnel plots and induced asymmetry on certain outcome scales (Sutton, 2009), Egger's test (Egger, Smith, Schneider, & Minder, 1997) was used to judge the symmetry of the funnel plot. Therefore, in this study, graphical (funnel plots) and statistical (Egger's test) methods were used to assess small study effects when sufficient (⩾10) studies were included in the meta-analysis (Higgins et al., 2019) and these two methods were provided by (Balduzzi et al., 2019).

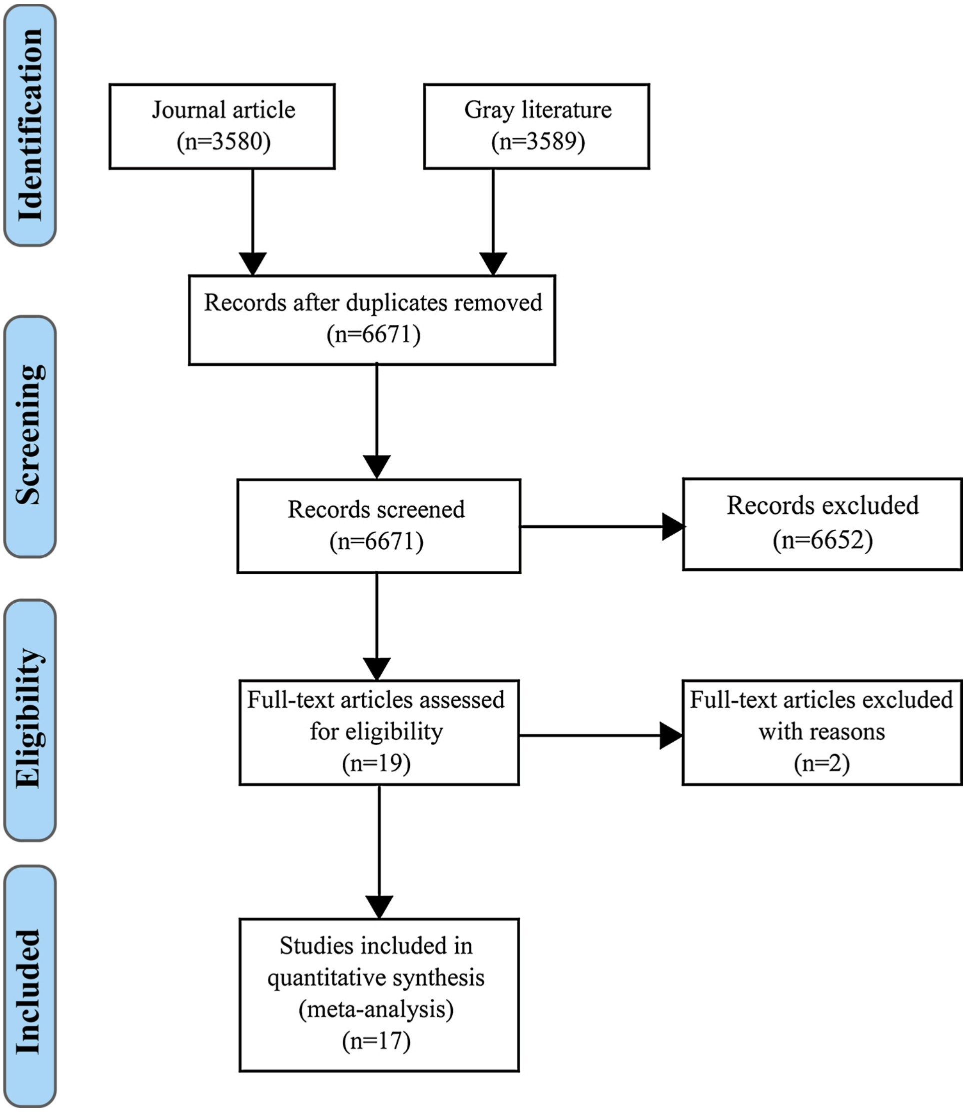

Fig. 1. Flow diagram of processing of records.

3. Results

3.1. Study selection

The process of article selection is shown in Fig. 1, based on the PRISMA flow diagram (Moher, Liberati, Tetzlaff, Altman, & group, 2009).

Fig. 1 shows the search process, wherein, after the eligibility criteria were applied and the records were screened, 19 remained. Of those, two additional articles were excluded. Since Hu et al. (2013b) used the same intervention and data as Hu et al. (2013a), these two studies actually comprised the same study. Second, the independent variable in the experimental group of Hui-jun (2010) included not only the physics concept map but also the teaching patterns. Therefore, the cause-and-effect relationships between physics concept map and students' scientific creativity could not be obtained. Thus, these two studies were excluded.

3.2. The categories of interventions

All interventions used in the studies included in this meta-analysis can be categorized into four categories according to the relationships between them and the cultivation of students' scientific creativity. Their details are as follows:Problem-solvingA series of cognitive procedures aimed at solving a problem, including a range of procedures from problem-recognition to solution, is often called problem-solving (Garrett, 1987) and is a fundamental component of human intelligence (Falomir & Olteţeanu, 2019). For example, rescheduling a trip due to unexpected circumstances, playing Tangram/origami, or finding a solution to an ill-structured problem (Newell & Simon, 1972) are well-known problem-solving tasks. As Mumford, Connelly, Baughman, and Marks (1994) stated, problem-solving involves the combination and reorganization of extant knowledge structures, similar to the mental process underlying students' scientific creativity. Therefore, students' scientific creativity can be regarded as a process of problem-solving (Getzels & Csikszentmihalyi, 1975). Following this view, an intervention that provides opportunities for students to solve science problems will support their scientific creativity. In this study, several studies improved students' scientific creativity by solving problems. For example, Gujarathi (1992) developed an intervention named *Training of Scientific Creativity* consisting not only of theory training regarding the steps of problem-solving, but also open-ended experiments requiring students to predict the hypothesis and identify various factors. Aktamis and Ergin (2008) developed an intervention named scientific process skills that included several necessary skills (define problems, observe, analyze, etc.) for solving science problems. Siew and Ambo (2018) developed an integrated project-based and STEM teaching and learning module that can expose students to real-world science problems. Besides these three studies, three further similar studies are included in this meta-analysis (Demirhan & Şahin, 2019; Ijjarilakmapura & Geetha, 2013; Saggu, 2012). In sum, problem-solving plays a central role in this type of intervention, with researchers training students to acquire problem-solving skills or providing students with opportunities to experience the problem-solving process. As was mentioned in Section 1, scientific problem-solving is one of the product dimensions of scientific creativity; therefore, problem-solving interventions can also be regarded as the cultivation of students' scientific creativity from the dimension named ***product***.Collaborative learningIn collaborative learning, group learning occurs by distributing participants' own thoughts and expertise, by listening to and elaborating on the views of others, and by the creative and shared knowledge construction of different thoughts and conclusions to reach common goals (Hämälä inen & Vähäsantanen, 2011; Roschelle & Teasley, 1995; Sawyer, 2017). As Arvaja, Häkkinen, and Kankaanranta (2008) stated, a common feature of collaboration is active and joint construction of shared understanding, meaning, knowledge, and expertise among the group or community. In other words, the main idea is that participants collaborate in a shared knowledge construction process (Dillenbourg, 1999; Kneser & Ploetzner, 2001), which allows for knowledge flow between different individuals. As stated above in Section 1, scientific creativity means the constrained stochastic behaviors of science knowledge; that is, it can be accurately modeled as a quasi-random combination process (Simonton, 2003) in which students' scientific knowledge is subjected to free, relatively unconstrained, or quasi-random recombination, with the aim of finding original and useful permutations to produce ***product*** (Campbell, 1960; Simonton, 2003, 2009, 2011). Evidently, the flow behaviors between different individuals' knowledge might lead to the accumulation of science knowledge, meaning that scientific knowledge that participated in quasi-random recombination increased or led to the improvement of the quantity of possible combinations of scientific knowledge, which increases the probability of obtaining original and useful permutations (Simonton, 2003; Turnbull et al., 2010). This means that all interventions (even those not called "collaborative learning") that create a free flow of knowledge between individuals might support students' scientific creativity. This has been demonstrated by several studies in this meta-analysis. For example, Joy (2014) investigated the effectiveness of the 5E learning cycle model, which provides students with opportunities to share their thoughts, scientific knowledge, and so on in group discussions and collaborative learning situations; Siew et al. (2017) used problem-based learning (PBL, also regarded as a form of collaborative learning), which creates a supportive environment based on group learning to support students' scientific creativity; and Siew and Chin (2018) developed a teaching and learning module by integrating PBL and collaborative learning to support students' scientific creativity. Two further studies also included this type of intervention (Kartika, Saepuzaman, Rusnayati, Karim, & Feranie, 2019; Wulansari, Rusnayati, Saepuzaman, Karim, & Feranie, 2019). In sum, in this type of intervention, students work collaboratively to share their understanding and scientific knowledge in group discussion to promote the flow of scientific knowledge between individuals, thus leading to improved scientific creativity. As stated above, promoting the flow of scientific knowledge between individuals actually improves students' scientific creativity by generating scientific work; therefore, this type of intervention, or so-called "collaborative learning," can also be seen as cultivating students' scientific creativity from the dimension named ***process***. Conceptual constructionAs mentioned in Section 1, students' scientific creativity is defined as the constrained stochastic behaviors of science knowledge that aim to produce ***product***, and students' science knowledge plays a central role in the form of free, relatively unconstrained, or quasi-random recombination of behaviors (Campbell, 1960; Simonton, 2003, 2009, 2011). Students' science knowledge refers to the formation of an organized and coherent conceptual structure comprising scientific concepts (Snyder, 2000). Therefore, theoretically speaking, the interventions that help students to construct discrete science concepts can improve their scientific creativity by increasing the scientific knowledge in quasi-random recombination. Three studies in this meta-analysis use this type of intervention. Nasiri (2000) showed the relationship between physics teaching and scientific creativity, with physics teaching aimed at acquiring knowledge of the basic logical concepts of three topics (heat, electrostatic, and electric current) using computer-assisted instruction (CAI). Aydin (2018) developed students' scientific creativity by acquiring the concepts of "cell division and genetics." Finally, Rasul, Zahriman, Halim, Rauf, and Amnah (2018) developed students' scientific creativity using STEM of smart communities modules, the first and basic phase of which is knowledge building. In sum, in this type of intervention, students' scientific creativity is cultivated through conceptual construction, leading to the accumulation of science knowledge. As noted above, this type of intervention is the same as collaborative learning, which improves students' scientific creativity by generating scientific work; therefore, this type of intervention called "conceptual construction" can also be seen as cultivating students' scientific creativity from the dimension named ***process***.Scientific reasoningScientific reasoning is often depicted as a process that seeks causes for observations of complicated science phenomena (Ding, 2018; Lawson, 2004) and is hypothetico-deductive in nature (Lawson, 2002, 2004). Historically, the causal models and theories that make sense of the natural world are built by scientists relying on hypothetico-deductive reasoning (Ding, 2018; Lawson, 2004), and, to show this process, Lawson (2004) used the case of Haler to answer the causal question, "how do returning salmon find their home stream?" This process includes identifying puzzling observations, generating plausible explanations (hypotheses), deducing inferences (predictions), and testing predictions using experiments. Of course, several cycles of

Table 1
Characteristics of studies in the meta-analysis.

ID	Study	Grade	Male/female	Country	Intervention	*N* (experimental group)	*N* (Control group)	Duration	Measurement instrument (self developed?)	Group	*g*	var (*g*)
1	Aktamis and Ergin (2008)	7	Miss value	Turkey	Scientific process skills education	20	20	12 weeks	No	Problem-solving	1.58	0.36
2	Aydin (2018)	8	27/33	Turkey	learning based on 5E learning cycle model	30	30	6 weeks	No	Conceptual construction	0.10	0.26
3	Demirhan and Şahin (2019)a	19–21 (ages)	18/70	Turkey	Unstructured modeling	25	10.5	6 weeks	No	Problem-solving	1.99	0.44
4	Demirhan and Şahin (2019)b	19–21 (ages)	18/70	Turkey	Semi-structured modeling	21	10.5	3 weeks	No	Problem-solving	1.99	0.46
5	Demirhan and Şahin (2019)c	19–21 (ages)	18/70	Turkey	Structured modeling	21	10.5	6 weeks	No	Problem-solving	0.70	0.39
6	Gujarathi (1992)	9	35/25	India	Training in scientific creativity	30	30	2.5 months	Yes	Problem-solving	1.60	0.30
7	Hu et al. (2013a)	7	51/46	China	Learn to think	54	43	2 years	Yes	Scientific reason	0.81	0.21
8	Ijjarilakmapura and Geetha (2013)	Miss value	Miss value	India	Biological science inquiry model	60	60	Miss value	Miss value	Problem-solving	1.61	0.21
9	Joy (2014)	9	Miss value	India	5E learning cycle model	156	156	40 min × 24 classes	Yes	Collaborative learning	0.91	0.12
10	Kartika et al. (2019)	11	Miss value	Malaysia	PBL using Scientific and Scientific Critical Worksheets (SSCW)	23	23	Miss value	Miss value	Collaborative learning	0.74	0.30
11	Lin et al. (2003)	7-11	Miss value	China	[a]	554	533	2 years	Yes	Scientific reason	0.91	0.06
12	Nasiri (2000)	10	Miss value	Iran	CAI[b]	92	92	3 months	No	Conceptual construction	0.20	0.15
13	Rasul et al. (2018)	7-9	147/183	Malaysia	STEM of smart communities modules	161	169	6 weeks	Miss value	Conceptual construction	0.23	0.11
14	Saggu (2012)	7	Miss value	India	Socio-constructivist approach	60	60	45 days	No	Problem-solving	1.40	0.20
15	Siew et al. (2017)a	6 (ages)	103/113	Malaysia	PBL[c]	36	72	6 weeks	Yes	Collaborative learning	0.48	0.21
16	Siew et al. (2017)b	6 (ages)	103/113	Malaysia	PBL[c]-NHT	72	36	6 weeks	Yes	Collaborative learning	0.43	0.21
17	Siew and Ambo (2018)	5	26/34	Malaysia	PjBL[d]-STEM	30	30	6 weeks	Yes	Problem-solving	1.68	0.30
18	Siew and Chin (2018)	6 (ages)	75/99	Malaysia	PBL[c]-CL	72	72	6 weeks	Yes	Collaborative learning	1.11	0.18
19	Wulansari et al. (2019)	11	Miss value	Malaysia	PBL using Scientific and Scientific Critical Worksheets (SSCW)	35	30	Miss value	Miss value	Collaborative learning	0.75	0.26
20	Yang et al. (2016)	5	Miss value	Taiwan, China	CIST[e]	20	24	16 weeks	Yes	Scientific reason	0.63	0.31

[a] CASE means Cognitive Acceleration through Science Education.
[b] CAI means Computer-Assisted Instruction.
[c] PBL means Problem-Based Learning.
[d] PjBL means Project-Based Learning.
[e] CIST means Creative Inquiry-based Science Teaching.

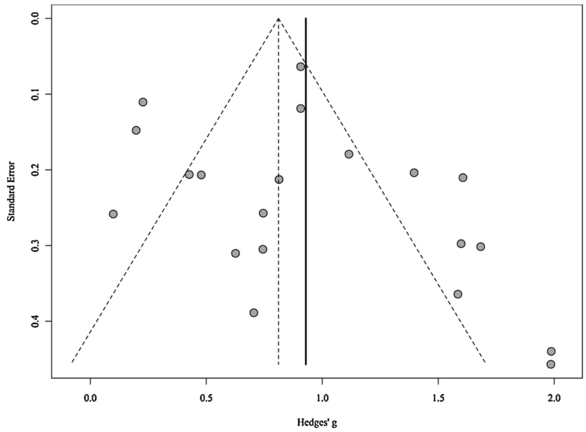

Fig. 2. Funnel plot of standard error by Hedges' *g*.

hypothetico-deductive reasoning might occur in this process due to the complexity of science phenomena (Lawson, 2002). As was stated in Section 1, seeking the causes for scientific phenomena is one of the products of scientific creativity; therefore, if an intervention provides opportunities for students to understand scientific phenomena in one or more cycles of hypothetico-deductive reasoning, then students' scientific creativity might be cultivated. Three studies in this meta-analysis developed interventions to improve students' scientific creativity based on cultivation of their scientific reasoning. Lin et al. (2003) used Cognitive Acceleration through a Science Education (CASE) program to improve students' scientific creativity. This CASE program stems from Piagetian psychology and is designed to accelerate students' development from concrete to formal operational thinking (Jones & Gott, 1998; Kind & Kind, 2007), the latter being hypothetico-deductive (Endler & Bond, 2008). Hu et al. (2013a) developed an intervention named "learn to think" (LTT) to improve students' scientific creativity by training their concrete and abstract thinking (Lawson, 2004). Yang, Lee, Hong, and Lin (2016) explored the effectiveness of creative inquiry-based science teaching (CIST) on students' scientific creativity, including activities that require students to generate more plausible explanations and predictions, to obtain the key independent variables influencing their experiments. In short, in this type of intervention, students' scientific creativity is cultivated by training their scientific reasoning. Students are actuated to generate more hypotheses or possible combinations of hypotheses, to generate corresponding predictions, and to test all predictions using experiments, seeking causes for complex science phenomena in one or more cycles of hypothetico-deductive reasoning. This type of intervention trains the fluency, flexibility, originality, and integrative and original synthesis of hypotheses and predictions, which form the ***traits*** dimensions of scientific creativity. Therefore, "scientific reasoning" can also be seen as cultivating students' scientific creativity from the dimension named ***traits***.

3.3. Study characteristics

The characteristics of the 17 studies included in this meta-analysis are shown in Table 1. Although 17 studies were included, 20 records are shown in Table 1. This is because there were two experimental groups, i.e., two interventions, in Siew et al. (2017), *problem based learning* and *problem based learning & cooperative learning*. Simultaneously, there were three experimental groups, i.e., three interventions, in Demirhan and Şahin (2019): *Unstructured modeling*, *Semi-structured modeling*, and *Structured modeling*. This gave a total of 20 interventions as recorded in the Table 1.

As shown in Table 1, the timespan of included studies ranged from 1992 (Gujarathi, 1992) until the end of 2019. However, most studies (13/17) were published after 2010, indicating that although the research on scientific creativity has attracted researchers' interest for a long time, only in recent years have researchers begun to develop interventions to improve students' scientific creativity.

Table 1 presents an overview of the variables grade, male/female, country, intervention, participants (*N*), duration of study, whether students' scientific creativity was measured by a self-developed instrument, category of intervention (group), *g* and var(*g*). Although some studies did not provide any information about students' grade or age, participants ranged from pre-school to university students. Most studies did not provide a gender breakdown of participants, but among those that did, the values of male/female were in the range of 0.76–1.41. The number of participants ranged from 40 to 1082, with large differences between studies. All participants came from one of five developing countries, China, India, Malaysia, Turkey, and Iran, which might indicate that the topic attracted more interest in developing than in developed countries. There were also significant differences in study duration, from weeks to years (some studies did not provide this information). Researchers also preferred to develop their own scientific creativity tests (eight of thirteen developed their own, four did not provide this information).

Table 1 also provides information on the interventions used in each experimental group. All interventions were categorized, eight as ***problem-solving***, six as ***collaborative learning***, three as ***conceptual construction***, and three as ***scientific reasoning***.

Table 2
Results of the powers of these four meta-analyses.

	Problem-solving	Collaborative learning	Conceptual construction	Scientific reasoning
$1-\beta$ err prob (no heterogeneity)	1.00	1.00	0.67	1.00
$1-\beta$ err prob (low heterogeneity)	1.00	1.00	0.54	1.00
$1-\beta$ err prob (moderate heterogeneity)	1.00	1.00	0.39	1.00
$1-\beta$ err prob (high heterogeneity)	1.00	0.9992	0.22	1.00

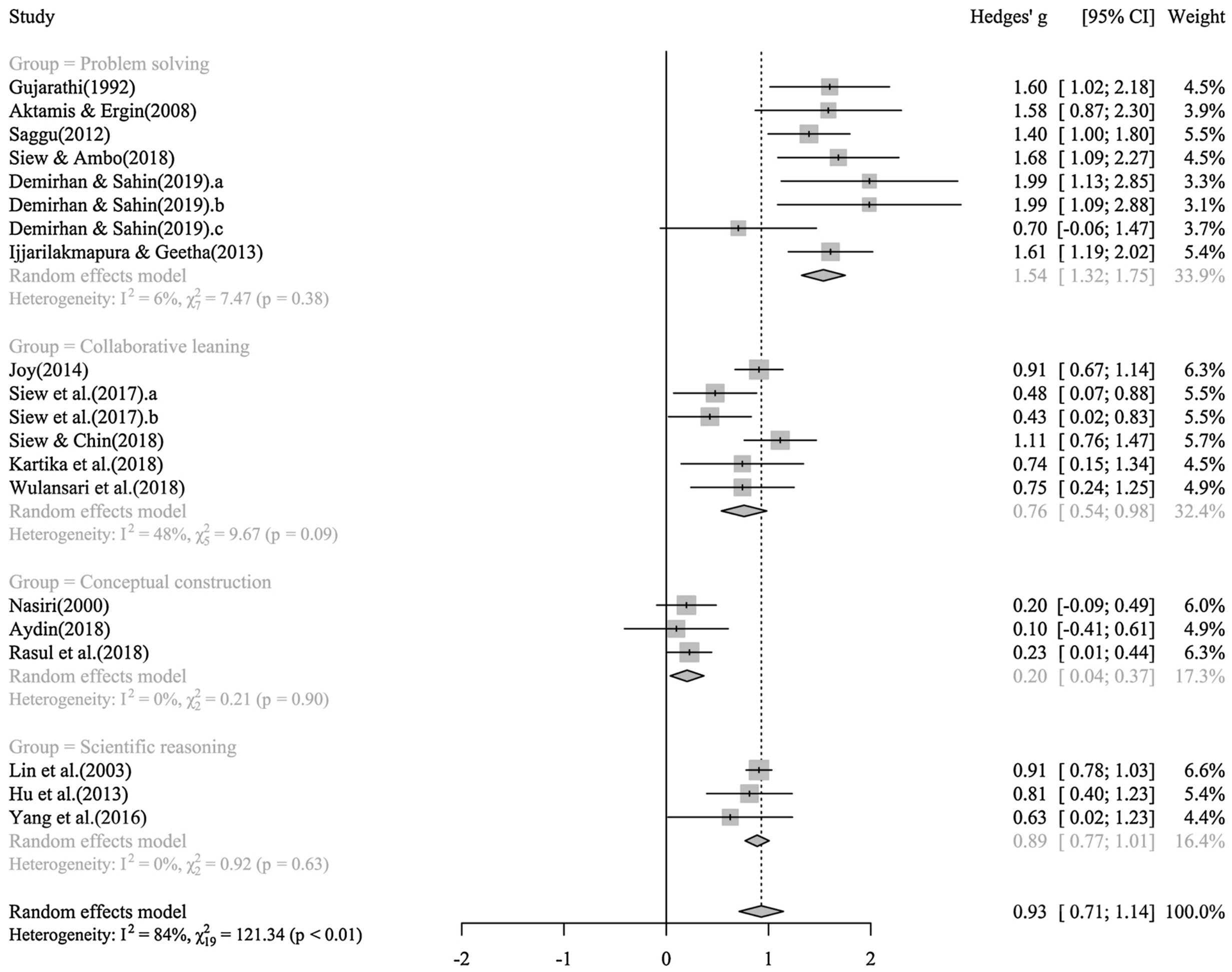

Fig. 3. Forest of results of studies and synthesis of subgroups.

3.4. Risk of bias across studies

To assess publication bias, a funnel plot was generated for the outcomes (see Fig. 2) and Egger's test was then conducted. The results (alternative hypothesis is asymmetrical in funnel plot) demonstrated that $t(18) = 1.12, p = .28$, suggesting that the effect sizes were not overestimated due to publication bias in this sample. That is, this meta-analysis was not affected by publication bias.

3.5. Risk of bias within studies

The risk of bias within studies is summarized in Table A.3 and the risk of bias graph is shown in Fig. A.4. Most studies had a low risk of *selection bias* and *reporting bias*. Only one study had a low risk of *blinding of participants* (Gujarathi, 1992). There is no study with a low risk of *blinding of personnel* or *detection bias*. Finally, there is no study with a high risk of *attrition bias*. In sum, though some research has low risk of bias across multiple dimensions (Gujarathi, 1992), the bias of all studies in this meta-analysis might lead to a larger effect size than the effect size actually is.

3.6. Results of individual studies and synthesis of results

The overall effect size of all interventions in this meta-analysis was 0.93 with a standard error of 0.11. The 95% confidence Interval was 0.71–1.14. The χ^2 test ($\chi^2_{19} = 121.34, p \leqslant 0.01$) and I^2 statistic ($I^2 = 84\%$) suggest substantial or considerable heterogeneity among the true effects of all interventions. Therefore, the results of interventions in different groups were synthesized, as shown in Fig. 3. The results of the powers of the four meta-analyses are also listed in Table 2.

As seen in Fig. 3, problem-solving had the greatest effect on students' scientific creativity ($g = 1.54; n = 8$), followed by scientific reasoning ($g = .89; n = 3$), collaborative learning ($g = .76, n = 6$), and conceptual construction ($g = .20; n = 3$). The χ^2 test and I^2 statistic showed a mild or no heterogeneity among the true effects of groups named *problem-solving*($\chi^2_7 = 7.47, p = 0.38; I^2 = 6\%$), *conceptual construction*($\chi^2_2 = 0.21, p = 0.90; I^2 = 0\%$), and *scientific reasoning*($\chi^2_2 = 0.92, p = 0.63; I^2 = 0\%$). Only the χ^2 test ($\chi^2_5 = 9.67, p = 0.09$) and I^2 statistic ($I^2 = 48\%$) suggest a moderate heterogeneity among the true effects of *collaborative learning*. Though Section 2.9 stated that the source of heterogeneity will be explained through meta-regression, Higgins et al. (2019) suggested that at least 10 observations should be available for each characteristic modeled to conduct meta-regression. There were only six interventions in the *collaborative learning* group of Joy (2014), Siew et al. (2017), Siew and Chin (2018), Kartika et al. (2019), and Wulansari et al. (2019).

Section 3.3 (see Table 1) shows the characteristics of the above five studies, which featured 312, 216, 144, 46, and 75 participants from India and Malaysia. That is, there is a large difference in the number of participants and countries between different studies. Furthermore, five studies (Kartika et al., 2019 and Wulansari et al., 2019 did not provide this information) used self-developed instruments. As Donker et al. (2014) stated, studies only using self-developed measures to evaluate student performance cause effect size inflation. In sum, there was good reason to believe that the source of heterogeneity in the *collaborative learning* group might stem from differences in study design. In other words, *collaborative learning* could produce a consistency effect on the cultivation of students' scientific creativity.

Table 2 shows the results of the power of meta-analysis. All syntheses of results except the synthesis of *conceptual construction* results have sufficient power to interpret the above results (suggested value is 0.8, see Higgins & Thompson, 2002) of the heterogeneity tests. This indicates that care should be taken when concluding the heterogeneity of the effectiveness of *conceptual construction*. Of course, this did not detract from concluding that the *conceptual construction* produced the smallest effect on students' scientific creativity compared to other groups.

In sum, problem-solving has a large effect on students' scientific creativity ($g = 1.54; n = 8$), as does scientific reasoning ($g = .89; n = 3$), while collaborative learning has a medium effect ($g = .76, n = 6$), and conceptual construction has a small effect ($g = .20; n = 3$). All four types of intervention could produce consistency effects on the cultivation of students' scientific creativity.

4. Discussion and conclusion

This meta-analysis investigated which interventions to improve students' scientific creativity are the most effective from preschool to university education. To establish the effects of interventions, we relied on studies for which usage of the intervention was assumed to cultivate students' scientific creativity. A search for literature published after 1950 yielded a total of 17 studies, including 20 interventions with 20 effect sizes. The studies' average mean effect size of Hedges' $g = .93$ (S.E. = 0.11), demonstrating that students' scientific creativity can indeed be improved by researchers' interventions.

Four types of intervention were included in this meta-analysis: *problem-solving*, *scientific reasoning*, *collaborative learning*, and *conceptual construction*. The effectiveness of different types of intervention can be ordered by the estimation of effect sizes (see Fig. 3): that is, *problem-solving* > *scientific reasoning* > *collaborative learning* > *conceptual construction*.

Of the interventions assessed, *problem-solving* was the most common and most effective ($g = 1.54$), with a focus on solving a series of problems. As Section 3.3 mentioned, this type of intervention cultivates students' scientific creativity from the dimension named ***product***; that is, interventions aimed at cultivating students' scientific creativity from the product dimension might be most effective.

Students' scientific creativity means the constrained stochastic behaviors of science knowledge. The *conceptual construction* and *collaborative learning* interventions were developed based on this view. As stated in Section 3.3, these types of intervention affect students' scientific creativity based on the ***process*** dimension, and compared to the other two types of intervention, they were less effective, perhaps indicating that interventions from the ***process*** dimension might be least effective. Additionally, *collaborative learning* ($g = .76$) is more effective than *conceptual construction* ($g = .20$). As Section 3.3 mentioned, *conceptual construction* improves students' scientific creativity by increasing the quantity of scientific knowledge in quasi-random recombination, while *collaborative learning* enhances the free flow of students' scientific knowledge, increasing not only the quantity of scientific knowledge in quasi-random recombination but also the quantity of possible combinations of scientific knowledge. This indicates that interventions aimed at enhancing the recombination behaviors between concepts belonging to conceptual structures might be more effective in developing students' scientific creativity. Of course, it cannot be concluded that the recombination behaviors between concepts belonging to conceptual structures are more important for students' scientific creativity than students' conceptual structure because the latter is a prerequisite for the former. It can, however, be concluded that if a student has held a conceptual structure, s/he can improve scientific creativity by enhancing the free flow between his/her own conceptual structure and those of others.

Scientific reasoning was the most effective intervention apart from *problem-solving*. As Section 3.3 mentioned, this type of intervention aims at improving students' scientific creativity from the "traits" dimension of scientific creativity. That is, this intervention produced a medium size effect on students' scientific creativity.

Table A.3
Risk of bias within studies in this meta-analysis.

ID	Study	Selection bias	Blinding of participants	Blinding of personnel	Detection bias	Attrition bias	Reporting bias
1	Aktamis and Ergin (2008)	1	1	1	1	− 1	− 1
2	Aydin (2018)	− 1	1	1	1	− 1	− 1
3	Demirhan and Şahin (2019)	1	1	1	1	− 1	− 1
4	Gujarathi (1992)	− 1	− 1	1	1	− 1	− 1
5	Hu et al. (2013a)	1	1	1	1	− 1	− 1
6	Ijjarilakmapura and Geetha (2013)	1	1	1	1	− 1	− 1
7	Joy (2014)	1	1	1	1	− 1	− 1
8	Kartika et al. (2019)	− 1	1	1	1	− 1	1
9	Lin et al. (2003)	− 1	1	1	1	− 1	1
10	Nasiri (2000)	− 1	1	1	1	− 1	− 1
11	Rasul et al. (2018)	− 1	1	1	1	− 1	− 1
12	Saggu (2012)	− 1	1	1	1	− 1	− 1
13	Siew et al. (2017)	− 1	1	1	1	− 1	− 1
14	Siew and Ambo (2018)	− 1	1	1	1	− 1	− 1
15	Siew and Chin (2018)	− 1	1	1	1	− 1	− 1
16	Wulansari et al. (2019)	− 1	1	1	1	− 1	1
17	Yang et al. (2016)	1	1	1	1	− 1	− 1

− 1 indicates a low risk of bias.
1 indicates high risk of bias.

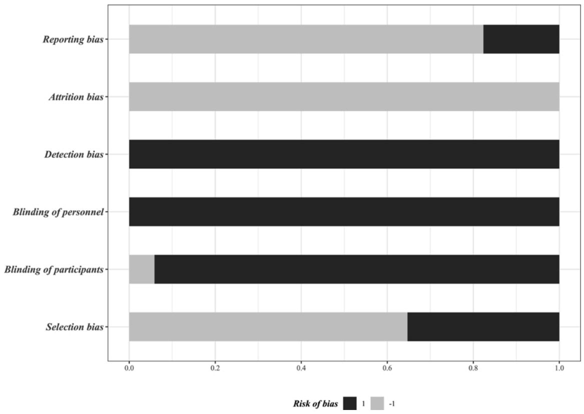

Fig. A.4. Risk of bias graph.

In sum, the interventions aimed at cultivating the product dimension of scientific creativity might be most effective, followed by the intervention based on training traits, and finally the intervention based on the process dimension.

Interpretation of our results is subject to several limitations. First, in this study, the electronic databases were selected carefully and the search strategy was designed by three researchers. While both journal articles and gray literature were included to obtain the final research sample, only studies written in Chinese and English were considered, potentially leading to publication bias. However, the result of a publication bias test indicated that this meta-analysis is unaffected (see Section 3.4).

Second, though this meta-analysis is unaffected by publication bias, many studies included have a high risk of bias in several dimensions (see Section A.3). As McMillan and Schumacher (2010) stated, in education, it is often difficult to meet the conditions of random assignment and manipulation (see Table 2.1). Simultaneously, in many studies, researchers play multiple roles in the research procedure, such as research design, conducting the research, and data analysis. Therefore, it is difficult to find a study with low risk of bias across all dimensions of educational research, and many studies with high risk of bias were included here. In many circumstances, risk of bias might lead to an effect size larger than the effect size actually is (see Table A.3). That is, the actual effects of the above four types of intervention might be smaller than the effects summarized in this meta-analysis. However, these risks of bias cannot prevent the conclusions made above from being obtained; that is, *problem-solving* is the most effective intervention used (g = 1.54), etc., because the focus of this meta-analysis is relative effectiveness.

Appendix A. Risk of bias within studies

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